

1. Complete the table by writing the constraint given as a written description using an inequality and set-builder notation.

Written Description	Inequality	Set-Builder Notation
All numbers greater than or equal to negative eight	$x \geq -8$	$\{x: x \geq -8\}$

2. A table of values that represents a function is shown.

x	-4	-1.25	0	1	3	6.25	10
y	8.2	6	5	4.2	2.6	0	-3

Use the table of values to determine the indicated key features for the function that is represented.

Key Feature	End Behavior	y -intercept
Description	As x -values increase, y -values <i>decrease</i> .	The y -intercept is <i>(0,5)</i> .

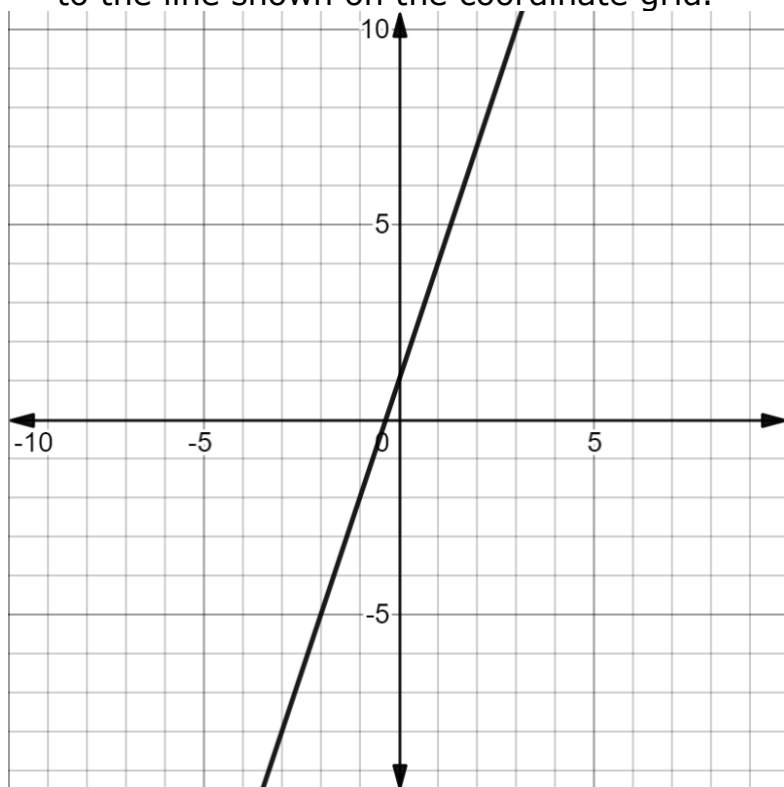
3. Write the equation of the line that goes through the point $(4, -7)$ and is perpendicular to the line $y + 6 = -\frac{2}{5}(x - 1)$. Use the table to show your work, if needed, to determine each form, and then give each form.

Work for Point-Slope Form	Work for Slope-Intercept Form
Point-Slope Form	Slope-Intercept Form
$y + 7 = \frac{5}{2}(x - 4)$	$y = \frac{5}{2}x - 17$

4. Write the equation of the line that is parallel to $x = 4$ and passes through $(-3, 7)$.

$x = -3$

5. Write the equation of the line in standard form that passes through (4,2) and is parallel to the line shown on the coordinate grid.



Show your work below

Standard form of parallel line

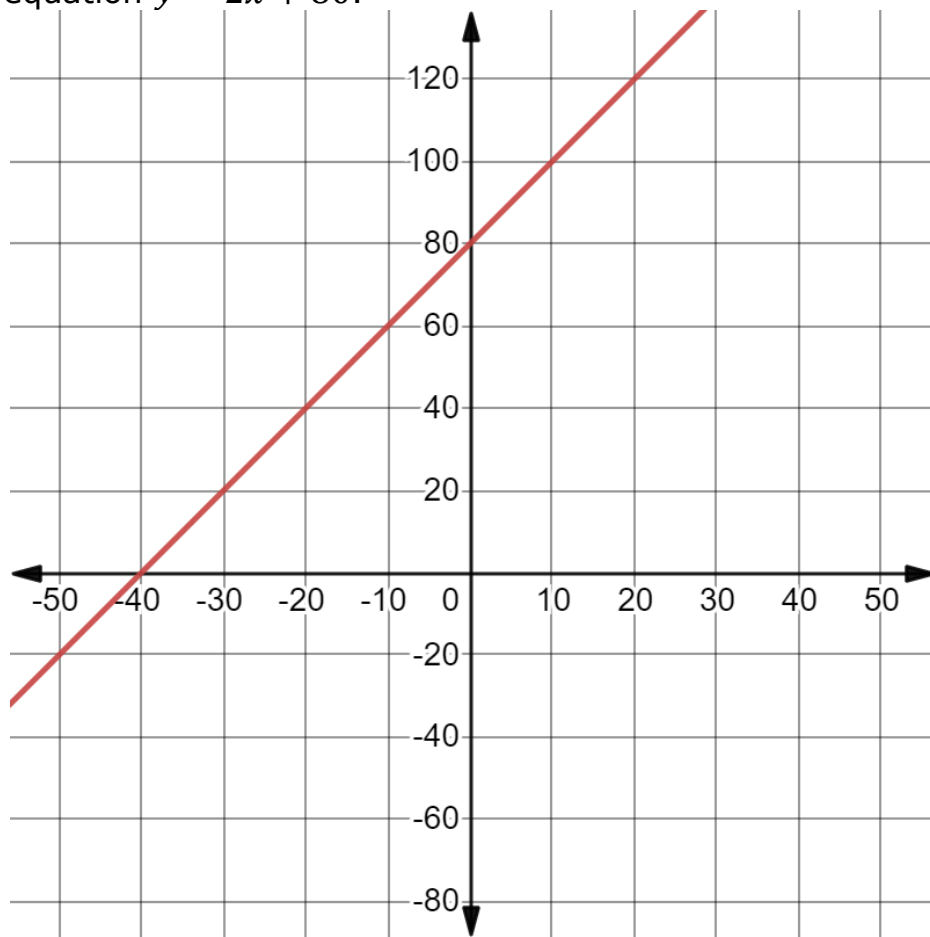
$$\underline{-3x + y = -10}$$

6. A research firm determined that the equation $y = 2x + 80$ can be used to model the total amount of spending by tourists in Florida, y , in billions of dollars, for the years 2013-2017 where x is the number of years since 2013.

- a. Determine the key features for the context. Show any necessary work below the table.

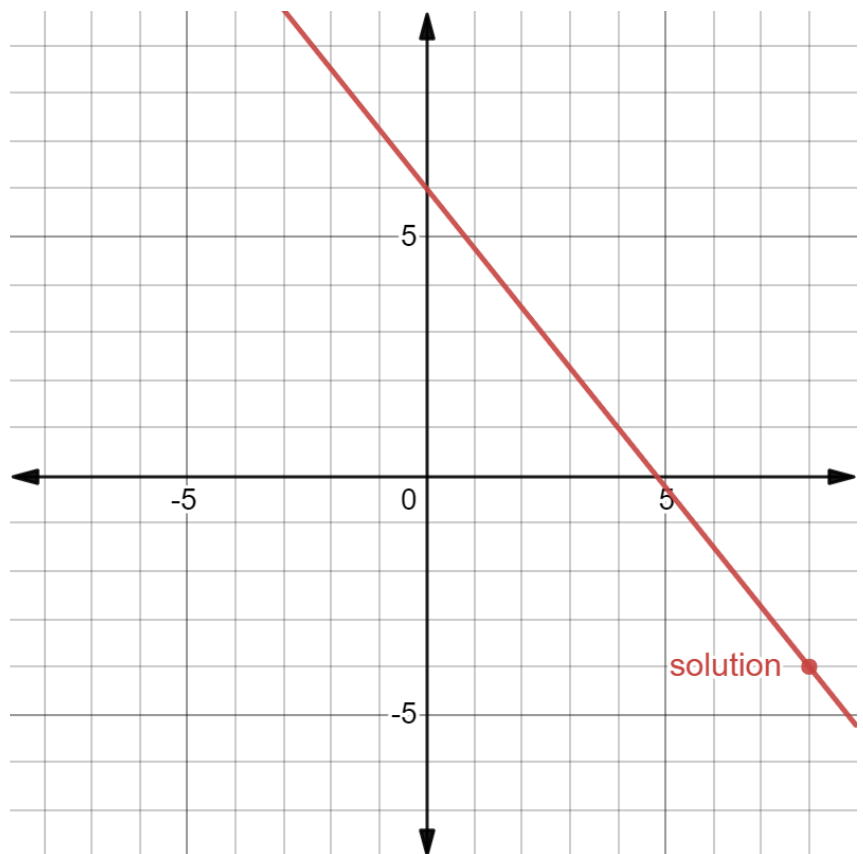
Domain	Range	As x -values increase, y -values ...
$0 \leq x \leq 4$	$80 \leq y \leq 88$	<i>increase</i>
Slope	x -intercept	y -intercept
2	$(-40, 0)$	$(0, 80)$

b. Graph the equation $y = 2x + 80$.



7. Graph the function $f(x) = -\frac{5}{4}x + 6$.
Label the point on the graph that is the solution to $f(x) = -4$ by writing "solution" next to it. Then, write the solution below.

Solution
$x = 8$



8. The function $S(t) = 1.084t + 17.392$ models the amount of money, S , in billions of dollars, the United States government spent on science, space, and technology for the time period 1999-2009, where t is the number of years since 1999.

a. Write the constraints for the variable t using set-builder notation.

Constraints $\{t: 0 \leq t \leq 10\}$

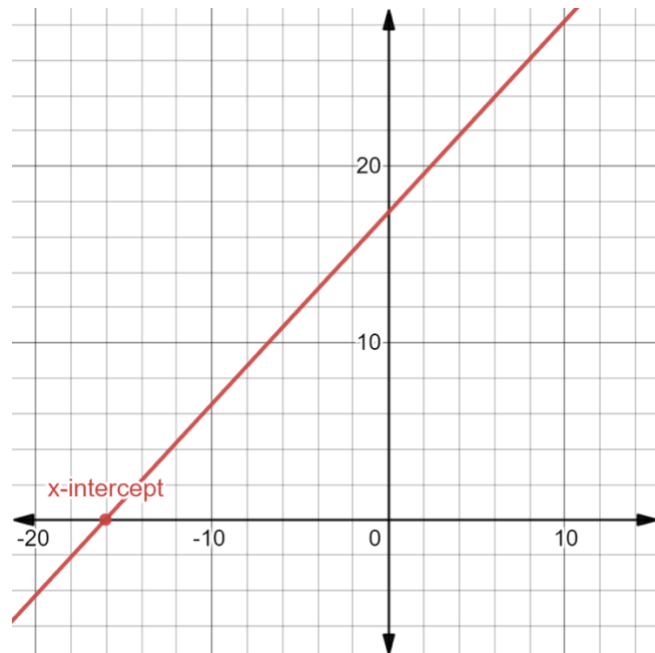
b. Determine the value of $S(7)$. $S(7) = 24.98$

c. Interpret the value of $S(7)$ in context.

Answers will vary: In 2006, US government spent 24.98 billions of dollars on science, space, and technology.

d. Name the key feature that is found when solving $S(t) = 0$. Then, label the key feature on the graph of $S(t) = 1.084t + 17.392$.

Key Feature x -intercept

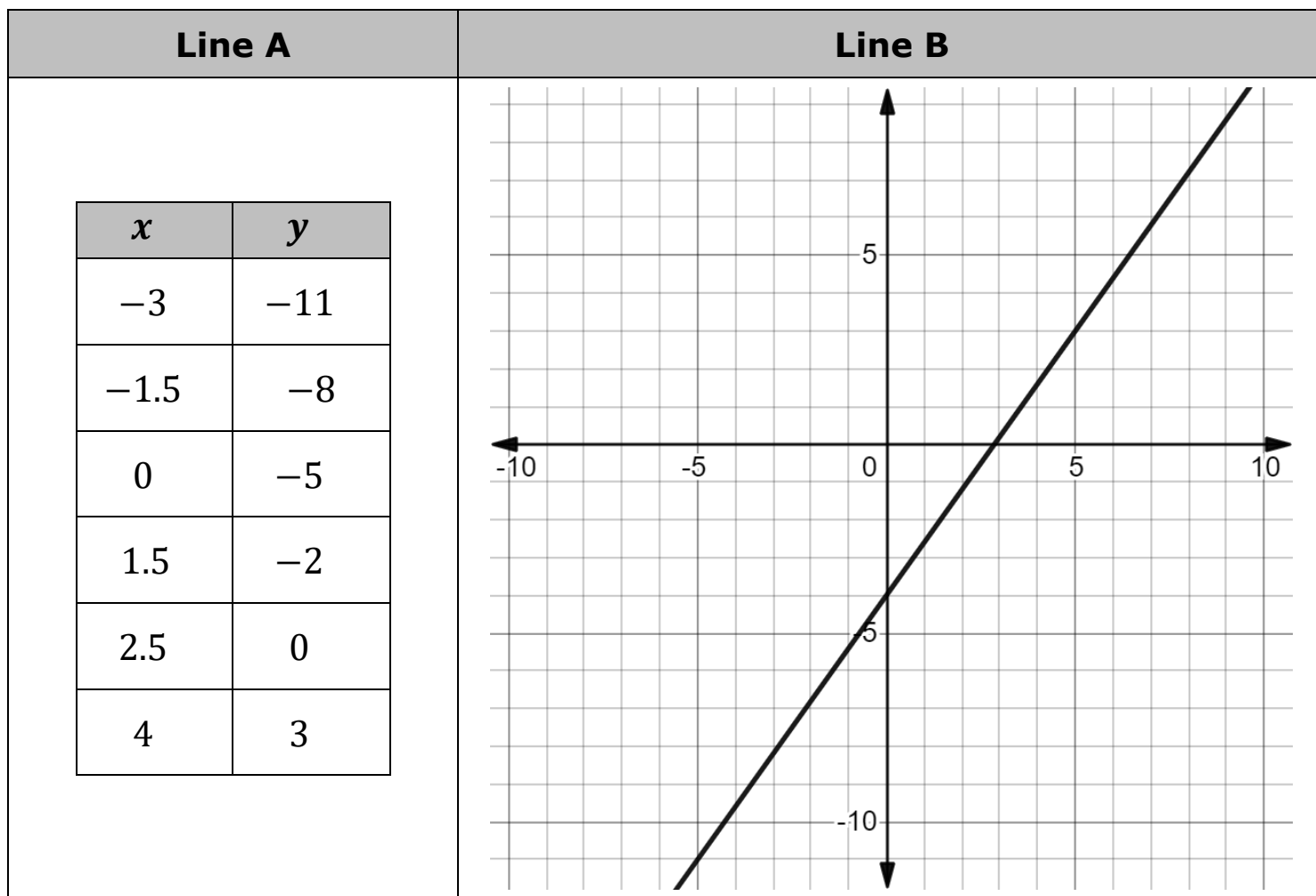


e. Determine if the value for $S(t) = 0$ is reasonable for the context. Explain why or why not.

Reasonable? ☐ No
☐ Yes

Explanation *Answers will vary: The x -intercept is approximately $(-16, 0)$, which means the year would be 1983 which is not within the given time period.*

9. Two different linear functions are shown.



Select the phrases that complete the comparison of the key features.

The x -intercept for

- ☐ line A
☒ line B

is greater than the x -intercept for

- ☒ line A.
☐ line B.

The range for line A is

- ☒ the same as
☐ different than

the range for line B.

The slope for

- ☐ line A
☒ line B

is smaller than the slope for

- ☒ line A.
☐ line B.